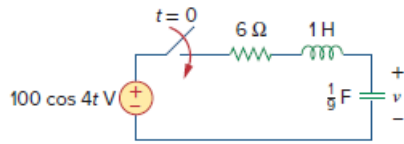


Problem

For the RLC circuit shown in Fig. 16.88, find the complete response if $v(0) = 100$ V when the switch is closed.

Figure 16.88:



Step-by-step solution

Step 1 of 8

Refer to Figure 16.88 in the text book.

Calculate the inductive reactance of the inductor in s domain.

$$\begin{aligned} X_L &= sL \\ &= s(1) \\ &= s \end{aligned}$$

Calculate the capacitive reactance in s domain.

$$\begin{aligned} X_C &= \frac{1}{sC} \\ &= \frac{1}{s\left(\frac{1}{9}\right)} \\ &= \frac{9}{s} \end{aligned}$$

Step 2 of 8

Since the capacitor is initially charged with 100 V.

Represent the initial condition of the capacitor with a voltage source of magnitude.

$$V(0) = \frac{100}{s}$$

Consider the input voltage source as,

$$v_i(t) = 100 \cos 4t$$

Take the Laplace transform.

$$\begin{aligned} V_i(s) &= \frac{100s}{s^2 + 4^2} \\ &= \frac{100s}{s^2 + 16} \end{aligned}$$

Step 3 of 8

Draw the equivalent circuit in Laplace domain.

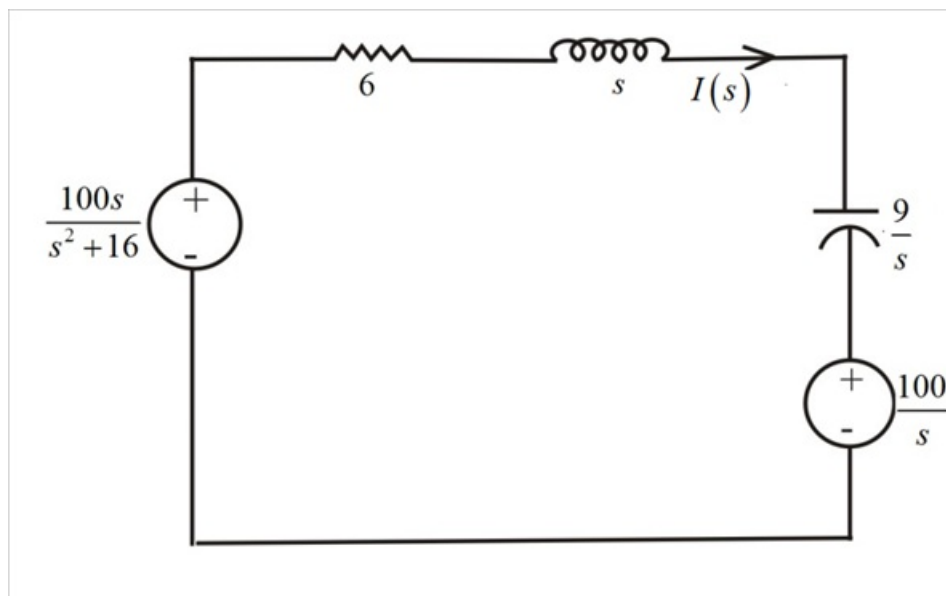


Figure 1

Step 4 of 8

Apply Kirchhoff's voltage law in the loop.

$$\begin{aligned}\frac{100s}{s^2+16} &= \left(6 + s + \frac{9}{s}\right) I(s) + \frac{100}{s} \\ \left(\frac{s^2+6s+9}{s}\right) I(s) &= \frac{100s}{s^2+16} - \frac{100}{s} \\ \left(\frac{s^2+6s+9}{s}\right) I(s) &= \left(-\frac{1600}{s^2+16}\right) \\ I(s) &= -\left(\frac{s}{s^2+6s+9}\right) \left(\frac{1600}{s^2+16}\right)\end{aligned}$$

Step 5 of 8

Determine the voltage response $v(t)$.

$$\begin{aligned}V(s) &= \left(\frac{9}{s}\right) I(s) + \frac{100}{s} \\ V(s) &= -\left(\frac{9}{s}\right) \left(\frac{s}{s^2+6s+9}\right) \left(\frac{1600}{s^2+16}\right) + \frac{100}{s} \\ V(s) &= \frac{100}{s} - \frac{14400}{(s+3)^2(s^2+16)} \dots\dots (1)\end{aligned}$$

Step 6 of 8

Express the term, $\frac{14400}{(s+3)^2(s^2+16)}$ in partial fraction.

$$\frac{14400}{(s+3)^2(s^2+16)} = \frac{A}{s+3} + \frac{B}{(s+3)^2} + \frac{Cs+D}{s^2+16} \dots\dots (2)$$

$$14400 = A(s+3)(s^2+16) + B(s^2+16) + (Cs+D)(s+3)^2$$

$$14400 = As^3 + 3As^2 + 16As + 48A + Bs^2 + 16B + Cs^3 + (6C+D)s^2 + (9C+6D)s + 9D$$

$$14000 = (A+C)s^3 + (3A+B+6C+D)s^2 + (16A+9C+6D)s + 48A+16B+9D$$

Step 7 of 8

Compare the powers of s .

$$A+C=0$$

$$3A+B+6C+D=0$$

$$16A+9C+6D=0$$

$$48A+16B+9D=14400$$

Solve for A, B, C and D .

$$A=138.24$$

$$B=576$$

$$C=-138.24$$

$$D=-161.28$$

Step 8 of 8

Rewrite equation (1) as:

$$\begin{aligned} V(s) &= \frac{100}{s} + \frac{138.24}{s+3} + \frac{576}{(s+3)^2} - \left(\frac{138.24s+161.28}{s^2+16} \right) \\ &= \frac{100}{s} + \frac{138.24}{s+3} + \frac{576}{(s+3)^2} - \frac{138.24s}{s^2+16} - \frac{161.28}{s^2+16} \\ &= \frac{100}{s} + \frac{138.24}{s+3} + \frac{576}{(s+3)^2} - \frac{138.24s}{s^2+16} - \frac{161.28}{4} \frac{4}{s^2+16} \\ &= \frac{100}{s} + \frac{138.24}{s+3} + \frac{576}{(s+3)^2} - \frac{138.24s}{s^2+16} - 40.32 \left(\frac{4}{s^2+16} \right) \end{aligned}$$

Apply inverse Laplace transform.

$$v(t) = (100 + 138.24e^{-3t} + 576te^{-3t} - 138.24\cos 4t - 40.32\sin 4t)u(t)$$

Therefore, the complete response of the circuit is

$$\boxed{v(t) = (100 + 138.24e^{-3t} + 576te^{-3t} - 138.24\cos 4t - 40.32\sin 4t)u(t) \text{ V.}}$$